About Simulation Output Governance Standard - (SOGS)

Simulation Results Actually Mean

A simulation may produce enormous amounts of data.

But producing an output is not the same as understanding what that output legitimately establishes.

A Simulation Output may be: direct, derived, aggregated, statistical, execution-specific, scenario-specific, or generated across millions of repeated runs.

The output may also be transformed, filtered, summarized, compared, or interpreted before someone eventually uses it to support a decision.

Simulation Output Governance Standard (SOGS) governs this transition.

Canonical Definition

Simulation Output Governance Standard (SOGS) defines the governance conditions under which outputs produced by Simulation Execution are identified, classified, bounded, interpreted, limitation-aware, documented, and connected to the execution basis that produced them so that direct Simulation Outputs remain distinguishable from derived results, aggregation, subsequent inference, and claims extending beyond the simulation itself.

Why SOGS Exists

Simulation programs often focus heavily on:

building the model

and:

running the simulation.

But a critical governance problem begins after execution:

What exactly does the result mean?

For example:

18 failures occurred in 250,000 simulation runs.

That is an output.

From that output, someone may conclude:

The system has a failure probability of 0.0072%.

That may already involve aggregation and interpretation.

Another person may then conclude:

The real system is 99.9928% reliable.

That is a much stronger claim.

These three statements are not methodologically equivalent.

SOGS exists to keep them separate.

Governed Space

Simulation Output Governance

SOGS governs Simulation Outputs after governed execution and before claims about Operational Reality, validation, or operational reliance are made.

The central question is:

What did the simulation actually produce, and how far may interpretation legitimately extend from that result?

Governance Flow

Governed Simulation Execution

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Simulation Output

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Output Identification

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Output Classification

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Execution & Scenario Binding

↓

Simulation Inference

↓

Output Limitations

↓

Output Documentation

↓

Governed Simulation Output

↓

Reality Correlation Output Is Not Inference

One of the most important distinctions in SOGS is:

Simulation Output ≠ Simulation Inference

For example:

Output: The simulated spacecraft entered Safe Mode in 14 executions.

Inference: Safe Mode activation appears associated with a specific simulated power and communication condition.

The inference may be supported by the output.

But it is still an inference.

SOGS keeps that interpretive step visible.

Simulation Inference Is Not Operational Reality

An even more important boundary is:

Simulation Inference ≠ Real-World Fact

A simulation may demonstrate:

Under these modeled conditions, this behavior occurred.

That does not automatically establish:

The real system will behave the same way.

SOGS therefore prevents a direct methodological leap from:

simulated behavior

to:

Operational Reality.

That next step requires downstream Reality Correlation and, where applicable, VALIDOS®.

Direct and Derived Outputs

SOGS distinguishes between:

Direct Simulation Output

and:

Derived Simulation Output.

A Direct Output may include: state reached, event occurred, trajectory produced, action selected, temperature exceeded, collision occurred, system recovered, or failure occurred.

A Derived Output may include: average, percentile, failure rate, distribution, ranking, comparison, aggregated metric, or post-processed result.

Derived Outputs remain legitimate.

But their relationship to the original executions must remain visible.

Aggregation Must Not Hide the Story

Large simulation programs often reduce millions of executions into a small set of numbers.

For example:

Mission Success Rate: 99.98%

That figure may be useful.

But SOGS asks:

Which executions were included?

Which failed?

Which were excluded?

Which Scenario Space was represented?

Did one rare but critical failure mode disappear inside the aggregate?

Was the output produced from nominal scenarios, stress scenarios, or the complete governed Scenario Set?

This protects organizations from allowing a convenient aggregate to conceal the conditions that produced it.

Outputs Must Remain Connected to Execution

A Simulation Output should remain traceable to:

Simulation Object

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Model & Environment

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Inputs

↓

Scenario

↓

Execution

↓

Output

Without this relationship, a result may later be interpreted outside the conditions under which it was generated.

SOGS therefore works directly with the execution lineage produced under SES.

Limitations Travel With the Output

A Simulation Output may inherit limitations from: assumptions, abstraction, fidelity, Input Data, scenario coverage, execution conditions, observation limitations, repetition sufficiency, model boundaries, or unresolved uncertainty.

These limitations do not disappear when an attractive chart or precise percentage is generated.

SOGS establishes:

Output processing does not erase upstream limitations.

Precision Does Not Equal Certainty

Simulation systems can produce extremely precise numbers.

For example:

0.00481763

The numerical precision may be mathematically correct.

But the result may depend upon: uncertain inputs, incomplete scenario coverage, simplified reality representation, assumptions, or limited execution conditions.

SOGS therefore distinguishes:

Computational Precision

from:

Strength of Interpretation.

Unexpected Outputs Matter

Simulation Outputs do not always confirm what was expected.

Unexpected behavior may reveal: a new interaction, a hidden dependency, an Edge Case, a Scenario Coverage Gap, an unstable model region, an execution anomaly, or a new simulation question.

SOGS therefore does not treat unexpected outputs as disposable noise by default.

They may create a feedback path:

Unexpected Output

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New Question

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Scenario Reassessment

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Additional Simulation Output Classification

Classification helps downstream users understand what type of result they are seeing.

A Simulation Output may be:

Direct

Derived

Aggregated

Intermediate

Final

Deterministic

Stochastic

Execution-Specific

Repetition-Based

Scenario-Level

Campaign-Level

or another governed class.

This becomes especially important when simulation outputs move between engineering teams, management, regulators, customers, or external partners.

Output Population Matters

An output should identify what population it represents.

For example:

Failure rate = 0.2%

means little unless it is known whether that represents: one scenario, one system configuration, one environment, one Repetition Set, an entire Scenario Set, or a complete simulation campaign.

SOGS keeps that population boundary visible.

Inference Boundary

A Simulation Inference should remain bounded by the simulation conditions that support it.

If a simulation examined:

Environment A

the output should not silently become evidence about:

Environments A, B, C, and D.

If a simulation represented:

one software version

the inference should not automatically extend to:

all future versions.

If the Scenario Set covered:

normal and moderate stress conditions

the conclusion should not automatically extend to:

extreme failure conditions.

Business Value

SOGS helps organizations answer questions that often become difficult after large simulation programs produce enormous amounts of results:

Which result came from which execution?

What is direct output and what is interpretation?

Which executions contributed to this aggregate?

Which results were excluded?

What limitations remain attached?

What can legitimately be claimed?

Where does the simulation stop and a reality claim begin?

This makes SOGS useful as both:

a Simulation Output governance framework

and:

a diagnostic layer for existing simulation programs. Use Case 1 — Space Mission Simulation OutputScenario

A space organization runs millions of simulations for autonomous spacecraft recovery.

The resulting dashboard reports:

Mission Recovery Success: 99.96%. Application

SOGS separates:

Direct Outputs — individual recovery outcomes.

Aggregated Output — 99.96% simulated recovery success.

Inference — recovery performs strongly across the simulated campaign.

The standard also preserves: which scenarios contributed, which runs failed, which executions were excluded, applicable power and communication conditions, scenario coverage limitations, and the execution lineage behind the aggregate.

Result

The organization can legitimately state what the simulation campaign demonstrated without converting:

99.96% simulated recovery success

into an unsupported claim that:

the real spacecraft has a 99.96% recovery probability. Use Case 2 — Autonomous Vehicle Safety SimulationScenario

An autonomous vehicle simulation reports:

99.99% successful collision avoidance

across a large simulation campaign. Application

SOGS examines whether the result represents: all Scenario Families, only nominal traffic, Edge Cases, degraded perception, adverse weather, combined conditions, and stress scenarios.

It also separates:

Observed Simulation Performance

from:

Inference about system safety. Result

The organization may conclude:

The system achieved 99.99% collision avoidance within the defined governed simulation population.

It may not automatically conclude:

The autonomous vehicle is 99.99% safe in Operational Reality.

That stronger claim requires downstream correlation and validation.

Relationship to SES

SES — Simulation Execution Standard governs what actually ran.

SOGS governs what those executions produced.

The handoff is:

Governed Simulation Execution Record

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Governed Simulation Output Relationship to SSS

SSS — Simulation Scenario Standard establishes the Scenario Space and Scenario Coverage.

SOGS ensures that output interpretation remains bounded by the scenarios actually represented.

Therefore:

Broad Output Volume ≠ Broad Scenario Meaning.

Relationship to Reality Correlation

SOGS deliberately stops before establishing whether simulated behavior corresponds to Operational Reality.

That next governance question is:

How strongly does the governed Simulation Output correlate with actual reality?

This belongs to the next SIMULOS® lifecycle space.

Relationship to VALIDOS®

VALIDOS® may subsequently evaluate whether a Simulation Output, Simulation Inference, reality-correlation claim, or simulation process can support a formal validation conclusion.

SOGS prepares the Simulation Output for that downstream assessment.

Relationship to INTEGROS®

SOGS deliberately does not create a separate Simulation Output Integrity governance space.

General integrity remains governed under INTEGROS®.

SOGS focuses specifically on:

Output Identification

Output Classification

Inference

Output Limitations

and:

Output Documentation.

Architectural Position

SOGS is the ninth Parent Standard of SIMULOS® — Simulation Governance Architecture.

The progression is:

Simulation Scenario Design

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Simulation Execution

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Simulation Output Governance

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Reality Correlation

SOGS therefore marks the transition from:

what happened inside the simulation

to:

what that result may legitimately support as an interpretation.

Why It Matters

Modern organizations increasingly have no shortage of Simulation Outputs.

They may have: millions of runs, enormous datasets, dashboards, distributions, AI-generated summaries, automated anomaly detection, and highly precise numerical results.

The real bottleneck can become:

meaning.

SOGS provides a governance layer for determining:

what the simulation actually produced, what was subsequently inferred, what remains limited, and where the claim must stop.

Foundational Principle

A Simulation Output becomes governable only when what the simulation directly produced remains distinguishable from how that output was processed, interpreted, limited, aggregated, and ultimately used to support claims beyond the execution that produced it.

Canonical Closing Statement

Simulation Output Governance Standard (SOGS) governs the transition from executed simulation to interpretable simulation result by ensuring that Simulation Outputs remain identifiable, classified, execution-bound, inference-bounded, limitation-aware, and documented. By separating direct output from derived output, aggregation from individual execution behavior, computational precision from interpretive certainty, and simulated behavior from claims about Operational Reality, SOGS enables organizations to use large and complex simulation outputs without allowing the meaning of those outputs to exceed the simulation conditions that actually produced them.

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